

E-COMMERCE SALES & CUSTOMER ANALYTICS

Project Documentation & Business Insights

Tools	Excel • PostgreSQL • SQL • Power BI
Dataset	50,000 e-commerce transactions • 2025
Objective	Analyze sales, profitability, customers, products, discounts and geography to generate actionable business insights.

This document summarizes the completed end-to-end analytics workflow, Power BI report and key business recommendations.

1. Executive Summary

The project analyzes 50,000 transactions from 2025 using Excel, PostgreSQL/SQL and Power BI. The objective is to understand revenue, profit, customer behavior, product performance, discount impact and regional performance, then translate those findings into business actions.

Metric	Result
Orders	50,000
Customers	8,604
Products	26
Units Sold	88,406
Revenue	■422.04M
Profit	■36.53M
Profit Margin	8.66%

Electronics: highest revenue category at about ₹262.49M, but only a 6.69% profit margin.

Fashion: strongest category margin at about 15.61%.

Discounts: 25% and 30% discount bands are loss-making in aggregate.

Customers: 6,951 repeat customers versus 1,653 one-time customers.

Geography: Maharashtra leads state revenue and profit.

2. Analytics Workflow

- Data validation: reviewed fields, values and transaction structure.
- PostgreSQL: loaded the dataset into the ECOMMERCE_SALES table in ecommerce_analytics.

- SQL analysis: completed KPI, sales, customer, product and business-insight analysis.
- Power BI: built five interactive pages: Executive Dashboard, Sales Trends, Product & Category Analysis, Customer Analysis and Geographical Analysis.
- Business interpretation: converted analytical patterns into recommendations.

3. Sales Performance

Quarterly performance improved toward year-end. Q4 was the strongest quarter, with approximately ₹114.44M revenue and ₹9.91M profit. Monthly analysis highlighted meaningful fluctuations that can be used for seasonal planning.

Quarter	Revenue	Profit	Margin
Q1	■101.36M	■8.78M	8.66%
Q2	■98.85M	■8.58M	8.68%
Q3	■107.39M	■9.26M	8.62%
Q4	■114.44M	■9.91M	8.66%

- Revenue and profit should be considered together because high sales do not automatically mean high profitability.
- The month-over-month analysis identifies growth and decline periods for management review.

4. Product & Category Analysis

Category	Revenue	Profit	Margin
Electronics	■262.49M	■17.57M	6.69%
Home & Furniture	■59.88M	■4.74M	7.91%
Fashion	■36.79M	■5.74M	15.61%
Sports	■29.27M	■3.63M	12.41%
Beauty	■18.87M	■2.67M	14.17%
Home & Kitchen	■14.74M	■2.18M	14.78%

- Laptop was the leading product in the Top 10 Products by Profit and also led the Top 10 Products by Revenue.
- Product decisions should use revenue, profit and margin together rather than sales volume alone.

5. Discount & Profitability Analysis

Discount	Orders	Revenue	Profit Margin
0%	6,034	■60.29M	17.05%
5%	8,918	■79.76M	13.47%
10%	12,531	■109.34M	9.60%
15%	9,991	■80.49M	5.88%
20%	7,431	■56.78M	2.18%

Discount	Orders	Revenue	Profit Margin
25%	3,573	■24.70M	-1.51%
30%	1,522	■10.68M	-5.44%

Key insight: profitability declines sharply as discounts increase. The 25% and 30% discount bands produce negative aggregate margins.

6. Customer Analysis

The customer analysis identified 8,604 customers: 6,951 repeat customers and 1,653 one-time customers. Repeat customers therefore represent about 80.78% of the customer base. The dashboard also ranks customers by revenue and profit, helping distinguish high-value customers from high-frequency but lower-value customers.

Recommended customer actions

- Use loyalty programs and personalized offers to strengthen repeat purchases.
- Identify high-value customers using revenue and profit, not order count alone.
- Run targeted re-engagement campaigns for one-time customers.

7. Geographical Analysis

Maharashtra is the strongest state in the analysis, generating approximately ₹74.30M revenue and ₹6.44M profit. The geographic dashboard also highlights leading cities such as Warangal, Nagpur, New Delhi, Siliguri and Mumbai.

State	Revenue	Profit	Margin
Maharashtra	■74.30M	■6.44M	8.67%
Uttar Pradesh	■42.53M	■3.67M	8.63%
Karnataka	■38.85M	■3.26M	8.40%
Gujarat	■34.30M	■2.94M	8.58%
Tamil Nadu	■31.41M	■2.80M	8.92%

- Use regional performance to guide inventory placement, marketing allocation and customer-acquisition strategies.

8. Power BI Dashboard Structure

Page	Purpose	Main Visuals
Executive Dashboard	Management overview	KPI cards, monthly trend, category, state/city, slicers
Sales Trends	Time-based analysis	Monthly/quarterly revenue and profit, category trends
Product & Category	Profitability analysis	Category revenue/profit/margin, discounts, top products
Customer Analysis	Customer value	Top customers, repeat vs one-time, customer table
Geographical Analysis	Regional performance	State/city revenue and profit, map, filters

9. Business Recommendations

Control excessive discounting:

Avoid broad 25–30% discounts unless product economics support them. Prefer targeted promotions and minimum-order thresholds.

Improve Electronics margins:

Review supplier costs, pricing, product mix and promotions because Electronics generates high revenue but a 6.69% margin.

Scale high-margin categories:

Increase attention to Fashion, Beauty and Home & Kitchen where margins are stronger.

Strengthen retention:

Use loyalty and personalized campaigns to convert one-time buyers and increase customer lifetime value.

Use profit with revenue:

Evaluate products, cities and categories using both absolute profit and margin.

Optimize regional resources:

Prioritize strong markets such as Maharashtra while testing growth strategies in weaker markets.

10. Technical Summary

- Database: PostgreSQL database ecommerce_analytics and table ECOMMERCE_SALES.
- SQL files: data validation, KPI analysis, sales analysis, customer analysis, product analysis and business insights.
- Power BI measures: Total Revenue, Total Profit, Total Orders, Total Customers, Total Products, Total Units Sold and Profit Margin.
- Interactive reporting: filters/slicers for dimensions such as state, category, payment mode and date.
- Final report: five Power BI pages designed for executive and detailed analysis.

11. Portfolio & Interview Summary

This project demonstrates an end-to-end Data Analyst workflow: data validation, relational database analysis, SQL querying, KPI validation, exploratory analysis, business interpretation, Power BI visualization and recommendation development.

30-second interview answer

“I built an E-Commerce Sales & Customer Analytics project using Excel, PostgreSQL, SQL and Power BI. I analyzed 50,000 transactions to understand revenue, profit, customer behavior, product performance, discounts and regional trends. A key finding was that Electronics generated the most revenue but had a lower margin, while 25% and 30% discounts were loss-making. I converted these findings into business recommendations and presented them through a five-page interactive Power BI dashboard.”

Next Portfolio Steps

- Upload SQL files, Power BI report, screenshots and documentation to GitHub.
- Create a professional GitHub README.
- Publish selected dashboard screenshots with the GitHub link on LinkedIn.
- Add the project to the Data Analyst resume and prepare project-based interview answers.